

**This assignment will not be graded and is only for practice.**

### Multiple Select Question (MSQ)

1) Consider a function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as:

**1 point**

$$f(x, y) = \begin{cases} \frac{2xy}{\sqrt{2(x^2+y^2)}} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

Choose the set of correct options.

- ☐  $|x| \leq \sqrt{x^2 + y^2}$  and  $|y| \leq \sqrt{x^2 + y^2}$ .
- ☐ If  $(x, y) \rightarrow (0, 0)$ , then  $f(x, y) \rightarrow 0$ .
- ☐  $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$  exists.
- ☐  $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$ .
- ☐  $f(x, y)$  is continuous at the origin  $(0,0)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$|x| \leq \sqrt{x^2 + y^2} \text{ and } |y| \leq \sqrt{x^2 + y^2}.$$

If  $(x, y) \rightarrow (0, 0)$ , then  $f(x, y) \rightarrow 0$ .

$\lim_{(x,y) \rightarrow (0,0)} f(x, y)$  exists.

$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$ .

2) Consider the following functions

**1 point**

$$f(x, y) = (\sqrt{1-x}, \sqrt{y})$$

$$g(x, y) = \sin((x^2 - 1)^2 + y^4)$$

$$h(x, y) = \frac{x}{x^2 + y^2}$$

$$s(x, y) = x^2 e^{y^2}$$

Which of the following is true?

☐ Domain of  $f = \{(x, y) \mid x \leq 1, y \geq 0\}$

☐ Domain of  $h = \mathbb{R}^2$

☐  $g \circ f(x, y) = \sin(x^2 - 1 + y^2)$

☐  $((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$

☐  $\lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of  $f = \{(x, y) \mid x \leq 1, y \geq 0\}$

$\lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$

3) Match the functions of two variables in Column A with their graphs given in Column B.

**1 point**

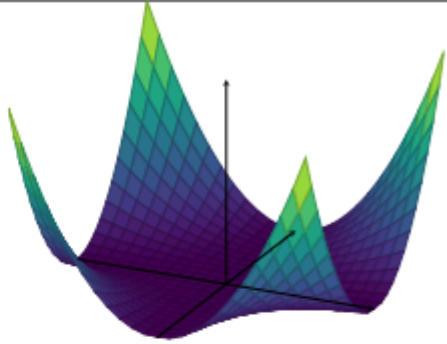
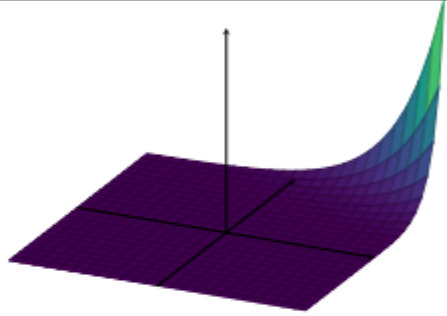
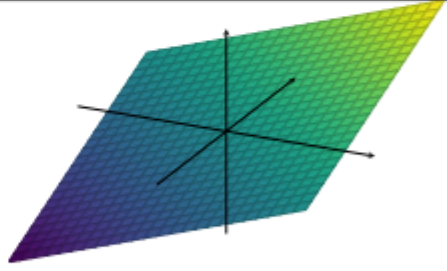
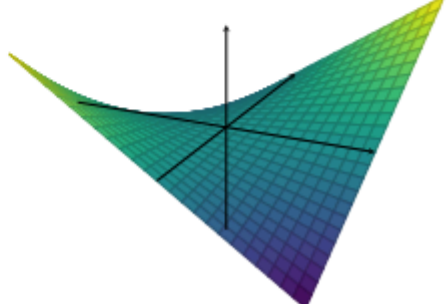
|      | Function of two variables<br>(Column A) |    | Graph of the function<br>(Column B)                                                  |
|------|-----------------------------------------|----|--------------------------------------------------------------------------------------|
| i)   | $f(x, y) = x + y$                       | 1) |    |
| ii)  | $f(x, y) = xy$                          | 2) |    |
| iii) | $f(x, y) = x^2y^2$                      | 3) |   |
| iv)  | $f(x, y) = 5^{2x+4y}$                   | 4) |  |

Table: M2W9P1

- ☐ i  $\rightarrow$  4, ii  $\rightarrow$  3
- ☐ i  $\rightarrow$  3, ii  $\rightarrow$  4
- ☐ iii  $\rightarrow$  1, iv  $\rightarrow$  2
- ☐ iii  $\rightarrow$  2, iv  $\rightarrow$  1

No, the answer is incorrect.

Score: 0

Accepted Answers:

i → 3, ii → 4

iii → 1, iv → 2

4) Consider two functions  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  and  $g : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as:

**1 point**

$$f(x, y) = (x^2, y^2)$$

and

$$g(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Which of the following option(s) is (are) true?

- ☐  $g \circ f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$
- ☐  $g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$
- ☐  $\lim_{(x, y) \rightarrow (0, 0)} g(x, y)$  exists.
- ☐  $\lim_{(x, y) \rightarrow (0, 0)} g \circ f(x, y)$  exists.
- ☐  $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g \circ f(x, y) = \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y)$
- ☐  $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g(x, y) = 1$
- ☐  $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = -1$
- ☐  $\lim_{(x, y) \rightarrow (1, 1)} g(x, y)$  does not exist.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$$g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g(x, y) = 1$$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = -1$$

## Numerical Answer Type (NAT)

5) Consider a function  $f(x, y) = xy$ . If the directional derivative of  $f$  at point  $(a, b)$ , in direction of  $(1, -1)$  and  $(1, 1)$  are 1 and 5 respectively. Then find the value  $\frac{2a+b}{\sqrt{2}}$ .

No, the answer is incorrect.

Score: 0

## Accepted Answers:

(Type: Numeric) 7

1 point

6) Consider the following statements:

**Statement 1:** Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function such that  $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f$  exists, where  $a, b \in \mathbb{R}$ . Then

$\lim_{(x,y) \rightarrow (a,b)} f$  exists.

**Statement 2:** Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function such that  $\lim_{y \rightarrow b} \lim_{x \rightarrow a} f = f(a, b)$ , where  $a, b \in \mathbb{R}$ . Then  $f$  is continuous.

**Statement 3:** Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function such that  $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f = \lim_{y \rightarrow b} \lim_{x \rightarrow a} f = f(a, b)$ , where  $a, b \in \mathbb{R}$ . Then  $f$  is continuous.

**Statement 4:** Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function such that  $f_x(a, b)$  exists. Then  $f$  is continuous.

**Statement 5:** Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function such that  $f_y(a, b)$  exists. Then  $f$  is continuous.

How many statements are correct?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: String) 0

1 point

7) Consider the following two functions  $f_1, f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as:

$$\begin{aligned} f_1(x, y) &= xy + 2x \\ f_2(x, y) &= x^2 + 2y + 3x^2y. \end{aligned}$$

Define a matrix  $A$  as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x}(3, 0) & \frac{\partial f_2}{\partial x}(1, 1) \\ \frac{\partial f_1}{\partial y}(3, 2) & \frac{\partial f_2}{\partial y}(\sqrt{\frac{7}{3}}, 2) \end{bmatrix}.$$

What will be the value of  $\det(A)$ ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -6

1 point

### Comprehension Type Question

An earthworm is crawling about on a sunny day. We draw coordinates axes on the ground where it is crawling and find that the body temperature of the earthworm at a point  $(x, y)$  is given by the function:

$$T(x, y) = 2x^2 + 3xy + y^2$$

Use this information to answer questions 8,9, and 10.

8) Which of the following option(s) is (are) true?

1 point

☐  $\lim_{(x,y) \rightarrow (1,1)} T(x, y) = \lim_{x \rightarrow 1} \lim_{y \rightarrow 1} T(x, y)$

☐  $T(x, y)$  is continuous at  $(1, 1)$

☐  $T(x, y)$  is a linear function.

☐  $T(cx, cy) = c T(x, y).$

- ☐ If the earthworm approaches the point  $(1, 2)$  from any direction, then the body temperature of the earthworm approaches 12 units.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\lim_{(x,y) \rightarrow (1,1)} T(x, y) = \lim_{x \rightarrow 1} \lim_{y \rightarrow 1} T(x, y)$$

$T(x, y)$  is continuous at  $(1, 1)$

If the earthworm approaches the point  $(1, 2)$  from any direction, then the body temperature of the earthworm approaches 12 units.

9) If the gradient of  $T(x, y)$  is  $(25, 18)$  at a point, then find the body temperature of the earthworm at that point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 77

**1 point**

10) Which of the following option(s) is (are) true?

**1 point**

- ☐ If the earthworm moves parallel to the  $X$ -axis, then rate of change of its body temperature at the point  $(a, b)$  is given by  $4a + 3b$ .
- ☐ If the earthworm moves parallel to the  $Y$ -axis, then rate of change of its body temperature at the point  $(a, b)$  is given by  $2a + 3b$ .
- ☐ If the earthworm moves parallel to the straight line joining the origin and the point  $(2, 3)$ , then rate of change of its body temperature at the point  $(1, 1)$  is given by  $\frac{29}{\sqrt{13}}$ .
- ☐ If the earthworm moves parallel to the straight line joining the origin and the point  $(0, 1)$ , then rate of change of its body temperature at the point  $(1, 2)$  is given by 10.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the earthworm moves parallel to the  $X$ -axis, then rate of change of its body temperature at the point  $(a, b)$  is given by  $4a + 3b$ .

If the earthworm moves parallel to the straight line joining the origin and the point  $(2, 3)$ , then rate of change of its body temperature at the point  $(1, 1)$  is given by  $\frac{29}{\sqrt{13}}$ .

Your score is: 0/10